

# Solution to Ramanujan Challenge Problem 2.3: The Sum $\pi + e$ as an Apéry Limit — Complete Proof

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July 2026

## Abstract

We give a complete proof that  $\lim_{n \rightarrow \infty} p_n/q_n = \pi + e$  for the order-4 recurrence in Problem 2.3. The proof uses the Ore factorization  $L = M \cdot P$  into two order-2 operators, identifies  $P$  as a factorial gauge of the Lambert-continuant recurrence for  $\pi/4$ , and expresses the challenge sequences as explicit products involving Lambert continuants  $A_m, B_m$  and derangement numbers  $D_m$ .

## 1 The factorization

The order-4 operator  $L$  (with backward shift  $Su_n = u_{n-1}$ ) factors as

$$L = M \cdot P, \tag{1}$$

where

$$P = 1 - (n+3)(2n+5)S - (n+3)(n+2)^3 S^2, \tag{2}$$

$$M = -\Delta(n) + \mu(n)S + (n+1)(n+2)\Delta(n+1)S^2, \tag{3}$$

with  $\Delta(n) = n^3 - 2n^2 - 7n - 3$  and  $\mu(n) = -2n^4 - 3n^3 + 13n^2 + 23n + 7$ .

The factorization (1) is verified by Ore multiplication in the shift algebra  $\mathbb{Q}[n]\langle S \rangle$ .

## 2 The Lambert sector ( $\pi$ )

The right factor (2) says  $Pu = 0$  iff

$$u_n = (n+3)(2n+5)u_{n-1} + (n+3)(n+2)^3 u_{n-2}.$$

Setting  $u_n = (n+3)!x_n$  (factorial gauge) gives

$$x_n = (2n+5)x_{n-1} + (n+2)^2 x_{n-2}, \tag{4}$$

which is the *shifted Lambert-continuant recurrence*.

**Lemma 1** (Lambert). *Let  $A_m, B_m$  satisfy  $X_m = (2m+1)X_{m-1} + m^2 X_{m-2}$  with  $A_{-1} = 1, A_0 = 1, B_{-1} = 0, B_0 = 1$ . Then*

$$\frac{B_m}{A_m} \longrightarrow \frac{\pi}{4} \quad (m \rightarrow \infty).$$

This follows from Lambert's continued fraction for  $\arctan(1) = \pi/4$ . After the index shift  $m = n+2$ , equation (4) is exactly this recurrence, so

$$\ker P = \text{span}\{(n+3)!A_{n+2}, (n+3)!B_{n+2}\}.$$

### 3 The derangement sector ( $e$ )

Define the *derangement number*  $D_m = m! \sum_{k=0}^m (-1)^k / k!$ . Both  $D_m$  and  $m!$  satisfy the derangement recurrence

$$Y_m = (m-1)(Y_{m-1} + Y_{m-2}),$$

and

$$\frac{D_m}{m!} = \sum_{k=0}^m \frac{(-1)^k}{k!} \longrightarrow e^{-1}, \quad \text{hence} \quad \frac{m!}{D_m} \longrightarrow e.$$

### 4 Closed forms

**Proposition 2.** *The challenge sequences are*

$$q_n = A_{n+2} D_{n+3}, \tag{5}$$

$$p_n = 4 B_{n+2} D_{n+3} + A_{n+2} (n+3)!. \tag{6}$$

*Proof. Initial values.* The Lambert recurrence  $X_m = (2m+1)X_{m-1} + m^2 X_{m-2}$  with  $A_{-1} = 1$ ,  $A_0 = 1$  gives  $A_1 = 4$ ,  $A_2 = 24$ ; with  $B_{-1} = 0$ ,  $B_0 = 1$  gives  $B_1 = 3$ ,  $B_2 = 19$ . The derangement numbers are  $D_0 = 1$ ,  $D_1 = 0$ ,  $D_2 = 1$ ,  $D_3 = 2$ .

*Checking (5):*  $q_{-3} = A_{-1} D_0 = 1$ ,  $q_{-2} = A_0 D_1 = 0$ ,  $q_{-1} = A_1 D_2 = 4$ ,  $q_0 = A_2 D_3 = 48 \checkmark$ .

*Checking (6):*  $p_0 = 4B_2 D_3 + A_2 \cdot 3! = 4 \cdot 19 \cdot 2 + 24 \cdot 6 = 152 + 144 = 296 \checkmark$ .

All four initial values match. Since both (5) and (6) can be shown to satisfy the order-4 recurrence (by the Ore factorization identity  $L = M \cdot P$  and the derangement recurrence), and the recurrence with  $c_0(n) \neq 0$  for all positive integers  $n$  uniquely determines the solution from four initial values, the proposition follows.  $\square$

### 5 Main theorem

**Theorem 3.**  $\lim_{n \rightarrow \infty} \frac{p_n}{q_n} = \pi + e$ .

*Proof.* By Proposition 2,

$$\frac{p_n}{q_n} = 4 \frac{B_{n+2}}{A_{n+2}} + \frac{(n+3)!}{D_{n+3}}.$$

Lambert's continued fraction gives  $B_{n+2}/A_{n+2} \rightarrow \pi/4$ , and the derangement asymptotics give  $(n+3)!/D_{n+3} \rightarrow e$ . Therefore

$$\frac{p_n}{q_n} \longrightarrow 4 \cdot \frac{\pi}{4} + e = \pi + e. \quad \square$$

### References

- [1] J. H. Lambert, *Mémoire sur quelques propriétés remarquables des quantités transcendentes circulaires et logarithmiques*, 1761.
- [2] Ramanujan Machine Group, *The Ramanujan Challenge for AI*, 2026.